Minimal Diagnostic Principles of Life: Criterion-Ablation and Observable Surrogate Selection in a Digital Ecosystem

Yuya Mukai

Mukai Entertainment
mukaiyuya@gmail.com

Abstract

All seven textbook biological criteria for life can be integrated as functionally interdependent processes within a single artificial life system. Here we implement cellular organization, metabolism, homeostasis, growth, reproduction, response to stimuli, and evolution (six at Level 4; evolution at Level 3) so that each criterion satisfies three conditions: sustained resource consumption, measurable degradation upon removal, and feedback coupling. We call this functional analogy.

Criterion-ablation experiments (n=30 per condition) show that disabling any single criterion causes statistically significant population decline (Holm-Bonferroni corrected, all $p \leq 0.004728$; Cliff's δ 0.39–1.00). Reproduction, response, and metabolism produce the strongest effects (Δ% of −91, −89, and −87, respectively). Pairwise ablations reveal sub-additive interactions consistent with shared failure pathways, and proxy controls provide evidence against tautological definitions.

A two-stage surrogate analysis then asks which observable measurements predict life-likeness without running the full ablation sweep. Phase 1 identifies six stable surrogates via Elastic Net stability selection. Phase 2 scales to 200 conditions across 5 environment regimes (14,000 runs), achieving out-of-regime test $R^2 = 0.950$ (permutation $p < 0.001$), which exceeds the single-predictor baseline of $R^2 = 0.921$.

Submission type: Full Paper
Data/Code available at: https://github.com/TheIllusionOfLife/minimal_life

Introduction

What distinguishes a living system from a merely complex one? Biology textbooks identify seven criteria: cellular organization, metabolism, homeostasis, growth and development, reproduction, response to stimuli, and evolution (Urry et al., 2020), one of several such lists in biology education, which we adopt as a working set, but artificial life (ALife) research has struggled to integrate all seven into a single computational system. This choice is pragmatic rather than exclusive: we use the seven criteria as an operational bridge across broader frameworks (NASA's Darwinian criterion (Joyce, 1994), autopoietic closure, and autonomy-centered minimal-life definitions), then test this bridge empirically via ablation. Most existing platforms implement a subset: evolutionary dynamics without metabolism (Ofria and Wilke, 2004), pattern formation without reproduction (Chan, 2019), or boundary self-organization without evolution (Plantec et al., 2023). Where criteria are nominally present, they often function as simplified proxies: static parameters or independent modules whose removal has no measurable effect on system behavior.

We argue that a meaningful computational model of life requires more than feature checklists. Each criterion must function as a functional analogy of its biological counterpart, satisfying three conditions:

1. Dynamic process: the criterion requires sustained resource consumption at every timestep, not a static lookup.

2. Measurable degradation: ablating the criterion causes statistically significant decline in organism viability.

3. Feedback coupling: the criterion forms at least one feedback loop with another criterion, precluding independent-module implementations.

This definition is system-agnostic: the three conditions test whether a subsystem fulfills the functional role of a life criterion, not whether it structurally resembles its biological counterpart. The same conditions can be applied to any ALife system with observable state variables and ablation toggles.

This definition operationalizes the intuition behind autopoiesis (Maturana and Varela, 1980), the chemoton model's integration of metabolic, boundary, and information subsystems (Gánti, 2003), and minimal-life frameworks (Ruiz-Mirazo et al., 2004) in a form amenable to experimental falsification.

We adopt a weak ALife stance: our system is a functional model of life, not a claim that the organisms are alive. The framework requires only: (1) per-criterion enable/disable toggles; (2) at least one measurable outcome variable; and (3) stochastic replication for statistical comparison. The contribution is methodological, comprising a two-stage design: Stage 1 integrates all seven criteria and verifies functional necessity via controlled ablation; Stage 2 applies criterion reduction (identifying which criteria contribute most to life-likeness) and surrogate selection (finding minimal observable measurements that predict life-likeness without running the full system) to the resulting 30-condition sweep.

The main contribution is a falsifiable experimental framework for testing functional necessity and interaction of life criteria in any ALife system. Concretely, this includes a hybrid swarm-organism architecture integrating all seven criteria as functionally interdependent processes; an operational functional analogy definition with three falsifiable conditions and a criterion-ablation methodology; pairwise ablation and proxy control experiments providing evidence against tautological definitions; and a powered surrogate analysis (200 conditions, 5 regimes, 14,000 runs) achieving out-of-regime generalization $R^2 = 0.950$.

Related Work

Table 1 surveys eight representative ALife systems across the seven criteria. Tierra (Ray, 1991) and Avida (Ofria and Wilke, 2004) achieve strong evolution but lack spatial organization, metabolism, and homeostasis. Polyworld (Yaeger, 1994) and Geb (Channon, 2001) integrate evolution with neural-network controllers and physical embodiment, but lack explicit metabolic or homeostatic subsystems. Stringmol (Hickinbotham et al., 2010) implements chemistry-based metabolism but omits homeostasis and spatial organization. Beer's Game-of-Life autopoiesis model (Beer, 2004) provides rigorous cellular organization but lacks evolution and reproduction. Lenia (Chan, 2019), Flow-Lenia (Plantec et al., 2023), ALIEN (Heinemann, 2008), and Coralai (Barbieux and Canaan, 2024) each cover additional subsets. Critically, none of these systems provides per-criterion ablation toggles enabling controlled removal experiments (Table 1, Abl. column); no existing system combines multi-step metabolism with active homeostatic regulation in an ablation-testable framework. Our framework operationalizes autopoiesis (Maturana and Varela, 1980; McMullin, 2004), the chemoton model (Gánti, 2003), and NASA's two-criterion definition (Joyce, 1994; Ruiz-Mirazo et al., 2004); open-ended evolution metrics (Bedau et al., 2000; Taylor et al., 2016) complement our criterion-ablation approach.

We acknowledge that Table 1 relies on self-assessment for our system; scores for other systems are our best reading of published descriptions. Independent cross-validation by system developers would strengthen these comparisons.

Table 1: Criterion comparison (five-level rubric; bold ≥4). Abl.=ablation-testable (per-criterion toggle enabling controlled removal). Rubric definitions in §S4 (Table S2).

System	CO	Me	Ho	Gr	Re	Rs	Ev	Σ	Abl.
Tierra (Ray, 1991)	1	1	1	1	3	2	4	13	–
Avida (Ofria and Wilke, 2004)	1	1	1	1	3	3	4	14	–
Polyworld (Yaeger, 1994)	3	2	1	2	4	4	4	20	–
Geb (Channon, 2001)	2	1	1	2	3	3	4	16	–
Stringmol (Hickinbotham et al., 2010)	2	3	1	1	3	1	3	14	–
Beer '04 (Beer, 2004)	4	2	2	1	1	2	1	13	–
ALIEN (Heinemann, 2008)	4	3	2	3	4	4	4	24	–
Ours†	4	4	4	4	4	4	3	27	✓

†Self-assessed. CO=Cell.Org, Me=Metab, Ho=Homeo, Gr=Growth, Re=Reprod, Rs=Response, Ev=Evol.

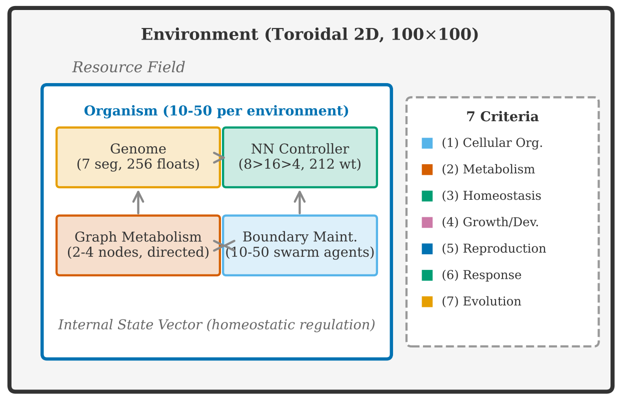

Figure 1: Two-layer architecture. Each organism comprises swarm agents maintaining a spatial boundary, a neural-network controller, a graph-based metabolic network, and a fixed 256-float genome organized into seven segments encoding all criteria. Organisms inhabit a continuous toroidal environment with a diffusing resource field.

System Design

The system implements a hybrid two-layer architecture (Figure 1). The outer layer is a continuous toroidal 2D environment (100×100 world units) containing a diffusing resource field. The inner layer consists of 10–50 organisms, each composed of 10–50 swarm agents that collectively maintain the organism's spatial boundary.

Seven Criteria

Table 2 maps each biological criterion to its computational implementation, ablation toggle, and feedback partners.

Full architecture, genome encoding (256-float vector organized into 7 segments; see §S4), neural-network specification (8→16→4 tanh; 212 weights), and parameter values appear in §S4.

Criterion-Ablation Experiment

This experiment tests whether each of the seven criteria is functionally necessary for organism viability, as predicted by the functional-analogy framework.

Protocol

For each of the seven criteria, we disable it while keeping all others active and compare against the fully enabled baseline (8 conditions total). A mid-run extension (ablation onset at step 1000) tests continuous necessity (§S1).

Outcome Measures

An organism dies when $b < 0.1$, $e \leq 0.0$, or age $> 20{,}000$ steps. The primary dependent variable is N_T (alive count at T=2000); secondary DVs are alive-count AUC and median organism lifespan. Metabolism, homeostasis, cellular organization, and response to stimuli are individual-level criteria; reproduction and evolution are population-level. Criterion-orthogonal controls (spatial cohesion, median lifespan) are reported in §S3.

Data Separation

Seeds 0–99 were used for calibration (threshold tuning); seeds 100–129 (n=30) are the held-out test set for all reported results. No ablation-condition outcomes were observed during calibration; thresholds targeted stable baseline populations only. All final experiments use the Graph metabolism engine.

Parameters. Each simulation runs for 2000 timesteps with population sampled every 50 on a 100×100 toroidal grid with 30 initial organisms (25 swarm agents each); resources regenerate at 0.01/cell/step; ablation toggles skip disabled processes without altering event ordering. Importantly, skipping a process does not redistribute saved energy or computation to other processes. The ablation simply removes the process's contribution, preserving the baseline resource budget. Run manifests lock parameters to reported results (registry file available in the supplementary repository).

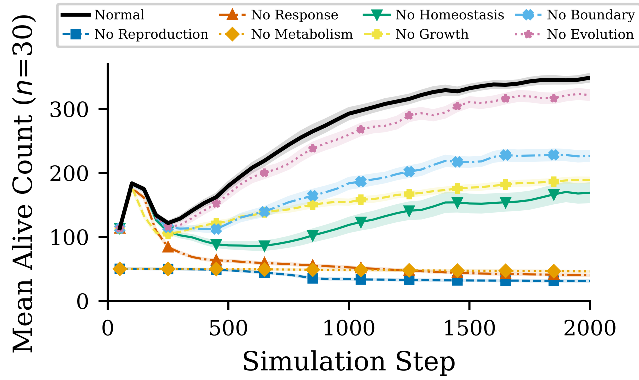

Figure 2: Population dynamics under criterion ablation. Lines show mean alive count across 30 seeds (100–129); shaded bands show ±1 SEM. Normal baseline (thick black) stabilizes near 349 organisms by step 1900. Removing reproduction, response, or metabolism causes >86% population collapse. Evolution ablation shows a modest 8% decline (Cliff's δ=0.39), consistent with optimization rather than short-term survival necessity.

Statistical Design

We test $H_1 : N_{\text{normal}} > N_{\text{ablated}}$ using the Mann-Whitney U test (Mann and Whitney, 1947) with Holm-Bonferroni correction (Holm, 1979) for seven simultaneous comparisons ($\alpha = 0.05$). Effect sizes are Cliff's δ (Cliff, 1993) and (for evolution-strengthening) Cohen's d, each with percentile bootstrap 95% CIs (2,000 resamples). The unit of replication is a seed; n=30 uses seeds 100–129; outcomes are N_T at T=2000 (primary) and alive-count AUC (secondary).

Results

All seven criterion ablations produce statistically significant population decline compared to the normal baseline (Table 3).

Three ablations cause near-total collapse (>86%): reproduction, response, and metabolism, the core viability loop (Figure 2). The key takeaway from Figure 2 is that the top three ablations converge to near-zero populations within 500 steps, while evolution ablation remains close to baseline, consistent with a clear effect hierarchy. Metabolic ablation collapses within 200 steps; reproduction ablation declines progressively; evolution ablation (δ=0.39) shows only modest impact, consistent with adaptation at longer timescales. Reproduction-ablated organisms live longer individually (median 751 vs. 242 steps) despite population collapse, confirming absent replacement rather than individual fragility (per-seed distributions in §S3).

Table 2: Mechanism specification: state variables, ablation operators, coupling pathways, and failure modes for each criterion. All criteria satisfy the three functional-analogy conditions (dynamic process, measurable degradation, feedback coupling).

Criterion	State Vars	Ablation	Coupling	Failure
Cell. Org.	$b \in [0,1]$	Skip repair; decay only	$e \to$ repair rate $\to b$	$b<0.1 \to$ death
Metab.	e, r, w; graph $\mathbf{p}$	Freeze (e, r, w)	throughput $\to e \to$ bdry repair	Energy depletion
Homeo.	$\mathbf{s} \in \mathbb{R}^3$	Skip NN delta; decay	$s_0 \to$ repair efficacy	State drift
Growth	$m \in [0,1]$; $\mathbf{d} \in \mathbb{R}^8$	Freeze m=0	$m \to$ bdry repair, sensing, metab. eff., reprod. gate	Reduced repair + sensing; no reprod.
Reprod.	Pop. event	Skip division check	energy cost $\to$ parent e	No replacement
Response	$\mathbf{v}$	Skip velocity delta	movement $\to$ resource $\to r$	Starvation
Evolution	genome $\mathbf{g}$	Copy w/o mutation	$\mathbf{g}$ variation $\to$ all	No adaptation

Table 3: Criterion-ablation results (n=30 per condition). Normal baseline mean: 348.9 (median: 349.5, IQR: 335.0–363.5). δ=Cliff's delta [bootstrap 95% CI]. Seven one-sided tests, Holm-Bonferroni corrected at $\alpha = 0.05$. $^{***}p < 0.001$, $^{*}p < 0.05$.

Condition	Mean	Δ%	δ [95% CI]	p_{corr}	Sig.
No Reprod.	31.2	−91.1	1.00 [1.00, 1.00]	<.001	***
No Response	40.0	−88.5	1.00 [1.00, 1.00]	<.001	***
No Metab.	46.1	−86.8	1.00 [1.00, 1.00]	<.001	***
No Homeo.	169.0	−51.5	1.00 [1.00, 1.00]	<.001	***
No Growth	188.8	−45.9	1.00 [1.00, 1.00]	<.001	***
No Boundary	226.7	−35.0	0.99 [0.98, 1.00]	<.001	***
No Evol.	321.8	−7.8	0.39 [0.11, 0.64]	.0047	*

Quantitative coupling evidence. Granger-style lagged F-tests (lags 1–5, within-seed Holm correction) show temporal precedence consistent with all three hypothesized pathways ($p_{\text{corr}} < 0.001$); energy and boundary integrity are strongly anti-correlated at lag 0 ($r = -0.78$, $p < 10^{-8}$). Intervention effects confirm non-modularity: metabolism ablation raises boundary by 41% (no repair energy) while response ablation reduces energy by 80% (organisms cannot reach resources); homeostasis ablation produces the largest internal-state change (−72%). Full coupling graph and intervention table appear in §S3.

Proxy Control Comparison

Three metabolism implementations on the same seeds (n=30), Counter (single-step, no waste), Toy (single-step with waste), Graph (multi-step network), all satisfy "dynamic resource consumption" yet produce qualitatively different dynamics (Figure 3): population size differs by 27% and genome diversity varies non-

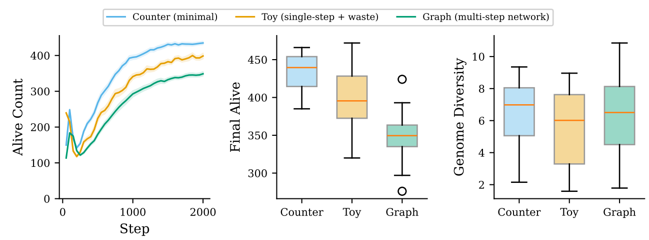

Figure 3: Proxy control comparison. Three metabolism implementations of increasing complexity on the same seeds (n=30). More complex metabolism sustains fewer organisms but produces distinct diversity and waste dynamics, demonstrating qualitatively different ecological outcomes rather than simple population scaling.

monotonically, indicating that the specific implementation shapes system behavior, providing evidence against tautological definitions. Importantly, the direction of ablation effects (metabolism removal causes population decline) is consistent across all three implementations, indicating that the qualitative conclusions are not implementation-specific.

Pairwise ablations (15 combinations; 6 reported in Table S3) show uniformly sub-additive synergy (bootstrap 95% CI excludes zero for 5/6 pairs), consistent with shared failure pathways rather than independent kill switches. The (metabolism, homeostasis) pair exemplifies convergent boundary failure: both routes reduce energy-funded boundary repair, so Δ_{AB}=298.1 falls below the additive expectation 450.7 (ratio 0.66). Full synergy formula, per-pair analysis, and the growth–reproduction interaction appear in §S5.

Table 4: Surrogate Analysis Summary (Phase 1 and Phase 2)

Analysis	Finding	Value
Phase 1 (30 conditions, 600 runs)		
PCA	Comp. $\geq$90% var.	4 (93%)
Stability (Enet)	Top criterion	Met. 0.678
Stability (Enet)	Bottom criterion	Evo. 0.374
Crit. Pareto	R^2 direction	All neg.
Cohen's d	full vs. all-off	10.68
Phase 2 (200 conditions, 14,000 runs)		
Stability sel.	Stable features	11/12
Train R^2	Permutation $p < 0.001$	0.954
Validation R^2	Seen regimes, new seeds	0.955
Test R^2	Unseen regimes	0.950
Baseline R^2	alive_auc only	0.921

Evolution at longer timescales. At 10,000 steps the evolution effect grows to d=1.42 (δ=0.66, $p < 0.001$), indicating that adaptation accumulates across generations; genome-drift divergence provides primary directional evidence (δ=1.00; see §S6); cyclic-recovery differences are mixed (p=0.341). See §S6 for full results and generation-stratified physiology.

Graded ablation ($\{1.0, 0.75, 0.5, 0.25, 0.0\}$ efficiency; n=30) shows a monotonic dose-response (Jonckheere-Terpstra, $p < 0.001$), consistent with metabolic contribution being quantitatively proportional rather than a binary switch. A state-neutral sham control (n=30) yields no significant difference ($p > 0.05$), validating that effects are functional. Full graded and cyclic environment results appear in §S11.

See Supplementary Material for mid-run extensions (§S1), ecology stressor protocol (§S2), phenotype clustering validation (§S3), pairwise ablation analysis (§S5), evolution timescale extension (§S6), full statistical results (§S7), failure mode analysis (§S8), fitness landscape evidence (§S9), threshold sensitivity and boundary topology robustness (§S10), graded/cyclic experiments (§S11), NN controller sensitivity (§S12), counter/toy engine comparison (§S13), and growth/reproduction separability (§S14).

Criterion Reduction Analysis

Protocol

A 30-condition ablation sweep (24 training, 6 held-out; 20 seeds each; 600 runs total) tests criterion-reduction feasibility. Conditions cover: the full 7-criteria baseline, the all-criteria-off control, 7 single-criterion drops, and 15 pairwise drops; 6 evolution-ablation conditions are held out. Decision thresholds are pre-registered in a decision rules document before held-out conditions are observed.

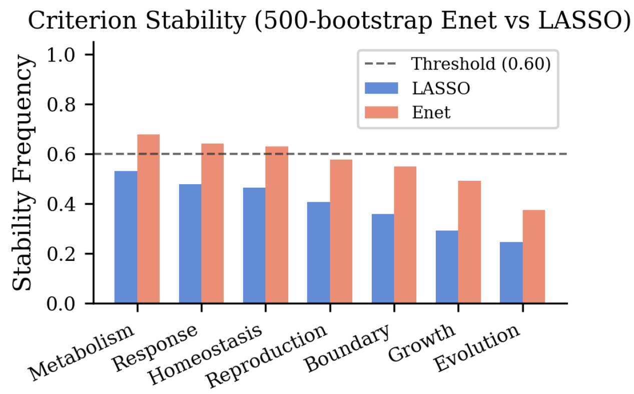

Figure 4: Criterion stability (500-bootstrap Enet vs LASSO; dashed line = 0.60 threshold). Enet ranks metabolism>response>homeostasis; evolution lowest.

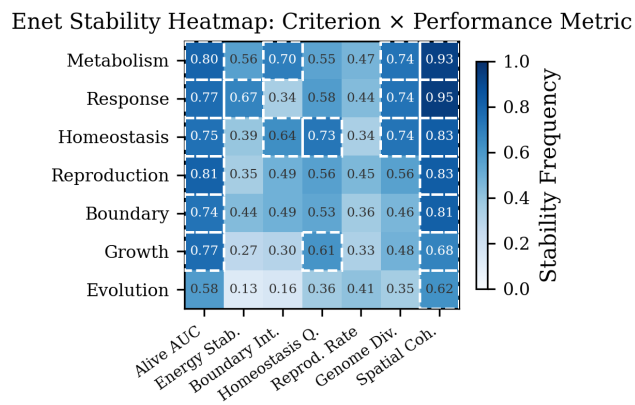

Figure 5: Enet stability heatmap (7 criteria × 7 performance metrics). Cells $\geq$0.60 are marked with dashed borders. Rows sorted by mean stability.

Results (Phase 1)

Enet bootstrapped stability scores (Meinshausen and Bühlmann, 2010) (500 subsamples, threshold 0.60) provide directional evidence: metabolism (0.678), response (0.641), and homeostasis (0.630) rank highest; evolution (0.374) and growth (0.492) rank lowest (Figure 4; per-metric breakdown in Figure 5). No criterion clears 0.60 under LASSO (maximum: metabolism=0.53). PCA requires 4 components for 93% variance (PC1=38%, PC2=31%); the biplot shows full-criteria and all-off at opposite poles (Figures 6 and 7). Forward Pareto selection yields all-negative R^2 values, a null result at this sample size (Cohen's d=10.68 validates the sweep design, but 24 training conditions are insufficient for predictive modeling). Enet stability rankings serve only as exploratory prioritization for Phase 2.

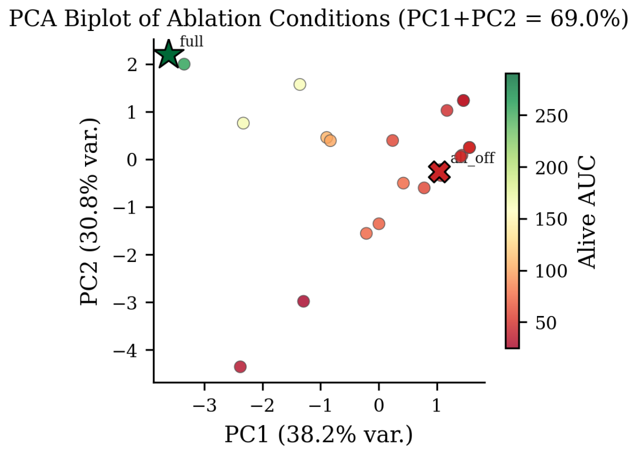

Figure 6: PCA biplot (PC1=38%, PC2=31%) of ablation conditions. Full-criteria and all-off sit at opposite poles; pairwise ablations cluster near all-off, reflecting multiplicative criterion interdependence.

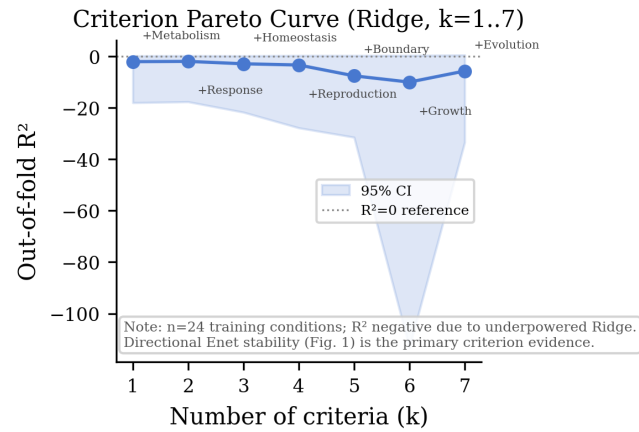

Figure 7: Criterion Pareto curve (Ridge, k=1..7). All $R^2 < 0$, indicating insufficient statistical power for predictive modeling at n=24 training conditions. Enet stability rankings serve as the primary directional evidence.

Observable Surrogate Analysis

A surrogate set is considered minimal-sufficient when it achieves ≥90% of the full-model R^2 on held-out regimes, with each retained feature surviving stability selection (frequency ≥0.60) and contributing non-redundant predictive variance.

Feature Engineering

From 14 scalar observables extracted from the 600-run sweep, VIF screening (threshold 10, following Montgomery et al. 2012) removes 8 collinear features, retaining 6: energy_autocorr (VIF=2.97), boundary_stability (1.32), homeostasis_var (1.64), spatial_cohesion_late (3.57), genome_diversity_late (9.59), and maturity_late (6.50).

Stability Selection

LASSO and Enet stability selection (500 subsamples, selection threshold 0.60 following Meinshausen and Bühlmann 2010, subsample ratio 0.5) both select all 6 retained features above threshold for all prediction targets (energy_autocorr: 0.998, boundary_stability: 1.00, homeostasis_var: 1.00, spatial_cohesion_late: 0.918–0.953, genome_diversity_late: 0.75, maturity_late: 0.877–0.984).

Pareto Curve

Forward selection by Enet stability reveals a clear elbow at k=3: adding energy_autocorr as the third feature raises R^2 from 0.007 to 0.165 (Figure 8). The full 6-feature set reaches $R^2 = 0.257$ (cross-validated, target: alive AUC). The best held-out prediction is regulation quality: $R^2 = 0.668$. The difference between alive-AUC ($R^2 = 0.257$) and regulation quality ($R^2 = 0.668$) reflects their diagnostic purposes: alive-AUC is a multi-criterion composite (harder to predict from few surrogates), while regulation quality captures a single criterion dimension (homeostatic control) that is more directly observable. Both serve diagnostic roles: alive-AUC for overall viability screening, regulation quality for homeostatic health assessment.

Data Leakage Mitigation

Aliased features (genome_diversity_late ↔ adaptation_score) are masked from their matching targets in all held-out evaluations.

Phase 2 Powered Analysis

The Phase 1 surrogate analysis is limited by 24 training conditions (600 runs). Phase 2 expands to 200 conditions across 5 environment regimes (14,000 runs total), with a 3-layer regime-based split: train (regimes A–C, seeds 0–39; n=4,800), validate (regimes A–C, seeds 40–69; n=3,600), and test (regimes D–E, seeds 0–69; n=5,600). The generalization axis is the environment regime: test evaluates transfer to unseen regimes (D–E); seeds 40–69 appear in both validation and test partitions to maximise power within each regime (a fully seed-disjoint split would require additional runs beyond current data). Decision thresholds are pre-registered before analysis.

From 12 candidate features (alive_auc, energy metrics, boundary_stability, demographic rates, and generation metrics), LASSO/Elastic Net stability selection (500 bootstraps, threshold≥0.60) retains 11 features (all frequencies≥0.984), replacing Phase 1 VIF screen-

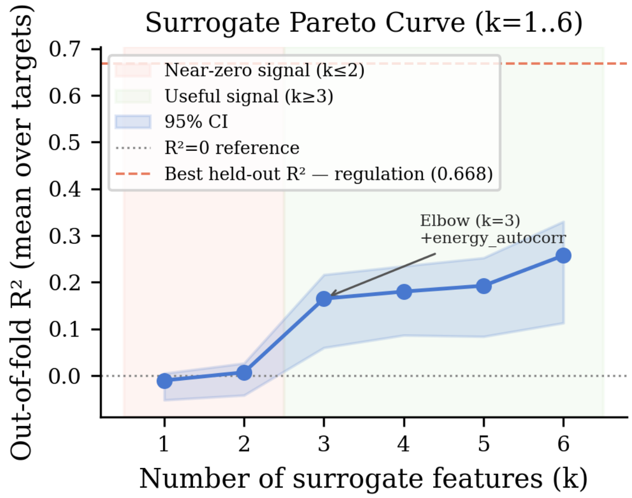

Figure 8: Surrogate Pareto curve (6 retained features, forward selection by Enet stability; target: alive AUC). Elbow at k=3 when energy_autocorr is added (R^2: 0.007→0.165). Dashed line: best held-out R^2 = 0.668 (regulation quality).

ing with implicit regularization of collinearity. The resulting model achieves train $R^2 = 0.954$, validation $R^2 = 0.955$, and out-of-regime test $R^2 = 0.950$ (permutation $p < 0.001$). The surrogate set exceeds the alive-AUC-only baseline (validation $R^2 = 0.921$), indicating multi-surrogate benefit (Table 4). Forward selection by LASSO coefficient magnitude shows that $k = 1$ surrogate (alive_auc) captures ≥90% of full-model validation R^2 ($R^2 = 0.921$ vs. 0.955 full); boundary_stability contributes the largest additional gain at $k = 4$ ($\Delta R^2 = +0.020$, $R^2 = 0.949$), with near-plateau thereafter. Excluding alive_auc, the remaining 11 non-alive_auc candidates achieve validation $R^2 = 0.941$ and test $R^2 = 0.845$ (LASSO, same protocol), indicating that predictive power remains substantial without the near-tautological alive-count proxy.

Discussion

Criterion interdependence. Single ablations reveal a hierarchy: reproduction, response, and metabolism form an essential triad (>86% decline), while growth and homeostasis occupy a middle tier (∼46%), followed by boundary (∼35%). Pairwise ablations show uniformly sub-additive interactions (Table S3), consistent with criteria converging on a shared viability attractor (boundary collapse), reachable via multiple independent failure routes rather than independent kill switches. This attractor interpretation separates emergent integration from by-construction necessity: if criteria were merely hardwired dependencies, ablation effects would be additive. The proxy control comparison further demonstrates that the specific implementation of a criterion shapes ecological dynamics (more complex metabolism sustains fewer organisms with distinct diversity profiles), providing evidence against tautological definitions.

Evolution at longer timescales. The modest 2,000-step evolution effect (d=0.78, δ=0.39) grows to d=1.42 (δ=0.66) at 10,000 steps (§S6), indicating that adaptation accumulates across generations; cyclic-environment recovery differences are mixed and treated as supplemental evidence (§S2). We explicitly acknowledge that our pre-registered H1 (directional energy selection) was not supported: generation-stratified energy trajectories are indistinguishable between evolved and non-evolved conditions (§S6). Primary evidence for adaptation is genome-drift divergence (δ=1.00, $p <$ 0.001) and population-level viability differences. A neutral-drift control (randomized parent selection with mutation active) and two-parent crossover experiment (segment-wise and uniform operators) provide additional evidence distinguishing genuine adaptation from drift (§S6). This places evolution at Level 3 (heritable variation, differential survival) rather than Level 4, and we acknowledge this limitation's impact on the seven-criteria integration claim.

Functional analogy across criterion types. Continuous criteria (metabolism, homeostasis) satisfy "sustained resource consumption" at every timestep; event-driven criteria (reproduction, evolution) maintain standing readiness (energy reserves, genetic variation), which constitutes functional engagement between discrete events.

Ablation side effects. Each ablation may induce secondary effects via shared state variables (e.g., disabling metabolism reduces energy, which impairs boundary repair and reproduction). We adopt knockout semantics: an ablation measures the total system-level consequence, inclusive of downstream cascades, analogous to gene knockouts. Off-target effect tables are in the supplementary materials.

Are criteria merely engineered? Four lines of evidence argue against this: (1) proxy control shows alternative implementations produce distinct dynamics (Figure 3); (2) pairwise ablations reveal sub-additive interactions inconsistent with independent modules (Table S3); (3) sham ablation control produces no significant difference; (4) graded ablation yields proportional dose-response (§S11).

Growth and criterion maturity. Growth contributes independently through boundary repair efficiency, sensing range, and metabolic throughput; §S5 confirms its effects extend beyond the reproduction maturity gate. Six criteria reach Level 4; evolution remains at Level 3. Achieving Level 4 would require open-ended dynamics or sustained directional trait change beyond our current scale.

What this does not show. This work does not establish that digital organisms are alive, does not demonstrate open-ended evolution, and does not model thermodynamic closure. The contribution is a falsifiable functional-analogy framework within a weak-ALife scope.

Criterion reduction. Phase 1 provides directional but not definitive evidence (§); Phase 2 resolves the power limitation (Table 4), with population viability as the dominant predictive axis and the multi-surrogate model adding 3.4 percentage points via metabolic, boundary, and demographic dimensions.

Limitations

Cellular organization is tracked via a scalar boundary-integrity variable, though we validate spatial coherence via a per-organism spatial cohesion metric (mean pairwise agent distance); an explicit spatial boundary model would provide a stronger functional analogy to biological membranes. Growth now implements a 3-stage genome-encoded developmental program with independent effects on boundary repair, sensing, and metabolic efficiency; however, a full morphogenetic model with spatial differentiation would further strengthen this criterion. Evolution reaches d=1.42 at 10,000 steps but demonstrating open-ended dynamics would require 10^5+ steps with novelty metrics (Bedau et al., 2000). Scale is limited to ∼300 organisms on a single machine (Mac Mini M2 Pro); 10× scaling is feasible with spatial partitioning, and larger populations might reveal emergent ecological phenomena. Generalization to regimes far outside the training distribution (e.g., extreme resource scarcity or population densities exceeding 1,000) remains untested. We adopt a weak ALife stance: this is a functional model, not a claim that digital organisms are alive.

Conclusion

The central finding is that none of the seven biological criteria is decorative. Removing any one causes statistically significant population decline ($p \leq 0.004728$, Holm-Bonferroni corrected; Cliff's δ 0.39–1.00), and pairwise ablations are uniformly sub-additive, pointing to shared viability pathways rather than independent kill switches. The surrogate analysis (Phase 2, out-of-regime $R^2 = 0.950$; Table 4) further suggests that a small set of observable measurements can approximate the full ablation sweep, supporting the notion of a minimal diagnostic principle of life.

Beyond the specific numbers, the functional-analogy framework and criterion-ablation methodology are portable: any ALife system with per-criterion toggles and observable state can be subjected to the same three-condition test. This makes the approach a reusable tool for the ALife community, not just a one-system result.

Code, data, and pre-registered decision rules are available at https://github.com/TheIllusionOfLife/minimal_life (software archived at https://doi.org/10.5281/zenodo.19351500; data archived at https://doi.org/10.5281/zenodo.18780935). Future work will scale to larger populations for open-ended evolution (Bedau et al., 2000; Taylor et al., 2016; Packard et al., 2019) and strengthen evolution toward Level 4 via extended runs with directional-selection metrics.

Acknowledgements

This work received no external funding.

Supplementary Material

Supplementary Material (§S1–§S14) is provided as a separate document.

References

Barbieux, A. and Canaan, R. (2024). Coralai: Intrinsic evolution of embodied neural cellular automata ecosystems. In The 2024 Conference on Artificial Life, ALIFE 2024. MIT Press.

Bedau, M. A., McCaskill, J. S., Packard, N. H., Rasmussen, S., Adami, C., Green, D. G., Ikegami, T., Kaneko, K., and Ray, T. S. (2000). Open problems in artificial life. Artificial Life, 6(4):363–376.

Beer, R. D. (2004). Autopoiesis and cognition in the game of life. Artificial Life, 10(3):309–326.

Chan, B. W.-C. (2019). Lenia: Biology of artificial life. Complex Systems, 28(3):251–286.

Channon, A. D. (2001). Passing the ALife test: Activity statistics classify evolution in Geb as unbounded. In Advances in Artificial Life, volume 2159 of LNCS, pages 417–426. Springer.

Cliff, N. (1993). Dominance statistics: Ordinal analyses to answer ordinal questions. Psychological Bulletin, 114(3):494–509.

Gánti, T. (2003). The Principles of Life. Oxford University Press, Oxford.

Heinemann, C. (2008). Artificial life environment. Informatik-Spektrum, 31(1):55–61.

Hickinbotham, S., Clark, E., Stepney, S., Clarke, T., Nellis, A., Pay, M., and Young, P. (2010). Specification of stringmol: A string-based artificial chemistry. Technical Report YCS-2010-458, Department of Computer Science, University of York.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics, 6(2):65–70.

Joyce, G. F. (1994). Foreword. In Deamer, D. W. and Fleischaker, G. R., editors, Origins of Life: The Central Concepts, pages xi–xii. Jones and Bartlett, Boston.

Mann, H. B. and Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other. The Annals of Mathematical Statistics, 18(1):50–60.

Maturana, H. R. and Varela, F. J. (1980). Autopoiesis and Cognition: The Realization of the Living. Boston Studies in the Philosophy and History of Science. Springer Netherlands.

McMullin, B. (2004). Thirty years of computational autopoiesis: A review. Artificial Life, 10(3):277–295.

Meinshausen, N. and Bühlmann, P. (2010). Stability selection. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 72(4):417–473.

Montgomery, D. C., Peck, E. A., and Vining, G. G. (2012). Introduction to Linear Regression Analysis. Wiley, 5th edition.

Ofria, C. and Wilke, C. O. (2004). Avida: A software platform for research in computational evolutionary biology. Artificial Life, 10(2):191–229.

Packard, N., Bedau, M. A., Channon, A., Ikegami, T., Rasmussen, S., Stanley, K. O., and Taylor, T. (2019). An overview of open-ended evolution: Editorial introduction to the open-ended evolution II special issue. Artificial Life, 25(2):93–103.

Plantec, E., Hamon, G., Etcheverry, M., Oudeyer, P.-Y., Moulin-Frier, C., and Chan, B. W.-C. (2023). Flow-lenia: Towards open-ended evolution in cellular automata through mass conservation and parameter localization. In The 2023 Conference on Artificial Life, ALIFE 2023. MIT Press.

Ray, T. S. (1991). An approach to the synthesis of life. In Langton, C. G., Taylor, C., Farmer, J. D., and Rasmussen, S., editors, Artificial Life II, volume XI of Santa Fe Institute Studies in the Sciences of Complexity, pages 371–408, Redwood City, CA. Addison-Wesley.

Ruiz-Mirazo, K., Peretó, J., and Moreno, A. (2004). A universal definition of life: Autonomy and open-ended evolution. Origins of Life and Evolution of the Biosphere, 34(3):323–346.

Taylor, T., Bedau, M., Channon, A., Ackley, D., Banzhaf, W., Beslon, G., Dolson, E., Froese, T., Hickinbotham, S., Ikegami, T., McMullin, B., Packard, N., Rasmussen, S., Virgo, N., Agmon, E., Clark, E., McGregor, S., Ofria, C., Ropella, G., Spector, L., Stanley, K. O., Stanton, A., Timperley, C., Vostinar, A., and Wiser, M. (2016). Open-ended evolution: Perspectives from the OEE workshop in York. Artificial Life, 22(3):408–423.

Urry, L. A., Cain, M. L., Wasserman, S. A., Minorsky, P. V., and Orr, R. B. (2020). Campbell Biology. Pearson, New York, 12th edition.

Yaeger, L. S. (1994). Computational genetics, physiology, metabolism, neural systems, learning, vision, and behavior or PolyWorld: Life in a new context. In Langton, C. G., editor, Artificial Life III, pages 263–298, Redwood City, CA. Addison-Wesley.